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Unit “ 4 : Session “ 11 : SLO “ 1 | Pipeline Hazards “ Choose the Best Answer

■ I. Multiple Choice

1. b) Data hazard “ dependence on a previous instruction's unfinished result.
2. b) Operand forwarding “ routes results directly from one stage to the next, bypassing the register file.
3. c) Control hazard “ the pipeline does not know which instruction to fetch until the branch resolves.
4. a) NOP (No Operation) “ inserting NOPs (bubbles) stalls the pipeline to handle hazards.
5. c) Structural hazard “ two instructions competing for the same functional unit at once.
6. b) Speculative execution (branch prediction) “ predicts the branch outcome and proceeds speculatively.
7. d) RAR (Read After Read) “ two reads never conflict; RAW, WAR and WAW are the true data hazards.
8. a) Increased CPI “ stalls add cycles, raising the average cycles per instruction.
9. a) Fetch “ the fetch stage suffers most, since the next PC is unknown until the branch resolves.
10. a) Duplicate functional units “ replicated hardware removes the resource conflict causing structural hazards.